

ABDELRAHMANE BENSAHLA

abdelrahmane4216@gmail.com | (201) 241-9488 | Jersey City, NJ | github.com/abdelrahmanebensahla | linkedin.com/in/abdelrahmanebensahla/

EDUCATION

Rutgers University - New Brunswick - Bachelor of Science in Computer Science - 3.2 GPA Sep 2021 – May 2025

WORK EXPERIENCE

New Jersey Turnpike Authority

May 2025 - Present

Full-Stack Engineer, Data Visualization & Dashboards Woodbridge, NJ

- Engineered a full-stack network visualization platform using Next.js, React, TypeScript, PostgreSQL, and Drizzle ORM, mapping 500+ enterprise devices and IP addresses and reducing network troubleshooting time by 40% across 3 departments.
- Designed and developed RESTful APIs and server-side endpoints with Node.js and Express.js, implementing full CRUD operations, authentication middleware, and pagination to serve real-time network topology data to frontend dashboards.
- Built and deployed a Python script management service that orchestrates automated network monitoring, data ingestion, and scheduled ETL pipelines, improving operational efficiency and eliminating 10+ hours of weekly manual processes.
- Architected normalized database schemas and implemented indexing strategies in PostgreSQL that optimized query performance by 60%, enabling sub-second load times for complex network topologies with 500+ nodes.
- Developed responsive, accessible UI components with Material UI and React, transforming complex network graphs into intuitive, interactive dashboards used daily by technical and non-technical stakeholders.
- Delivered end-to-end features through the complete SDLC: requirements gathering, database design, API development, frontend implementation, testing, and deployment within an Agile/Scrum team of 8+, maintaining a 95% sprint completion rate.
- Presented technical demonstrations to senior leadership, translating complex data insights into actionable recommendations that informed infrastructure planning and resource allocation decisions.

TECHNICAL PROJECTS

Enterprise Network Dashboard

Next.js | React | TypeScript | PostgreSQL | Node.js | Material UI | Drizzle ORM | REST APIs

- Designed and deployed a production-grade full-stack web application enabling real-time monitoring and management of 500+ enterprise network devices across NJTA infrastructure.
- Implemented a normalized PostgreSQL schema with Drizzle ORM and built optimized pre-configured queries, reducing average data retrieval time by 60% and supporting sub-second dashboard rendering.
- Built an intuitive CRUD interface with role-based access control (RBAC), dynamic filtering, and responsive data tables, adopted for daily use by network operations teams.

Job Fraud Detection Using Neural Networks & TF-IDF

Python | TensorFlow | Scikit-Learn | NLP | TF-IDF | Neural Networks

- Developed a deep learning classification model achieving 98% accuracy (vs. 72% baseline) on 10,000+ job postings, leveraging TF-IDF vectorization and class-weight balancing to handle severe data imbalance.
- Reduced false positives by 35%, potentially saving hiring teams 100+ hours of manual review through automated fraud detection.

AI Trajectory Planning with A* Algorithms

Python | A Algorithms | Binary Heaps | Performance Optimization*

- Implemented Forward, Backward, and Adaptive A* algorithms with a 30% performance improvement over baseline, optimizing memory usage via binary heaps and Manhattan heuristics for real-time pathfinding on 100×100 dynamic grids.

SKILLS

Languages: JavaScript, TypeScript, Python, SQL, Java, C, C++, HTML, CSS3

Frontend: React, Next.js, Redux, Material UI, Tailwind CSS, Responsive Design, Server-Side Rendering (SSR)

Backend: Node.js, Express.js, Django, FastAPI, REST APIs, GraphQL, Microservices, OAuth/JWT Authentication

Databases: PostgreSQL, MySQL, MongoDB, Redis, Drizzle ORM, Prisma, Database Design & Optimization

DevOps & Cloud: Docker, AWS (EC2, S3, Lambda, RDS), GCP, Azure, CI/CD, GitHub Actions, Git, Linux, Terraform

Testing & AI/ML: Jest, TDD, TensorFlow, PyTorch, Scikit-Learn, NLP, LLMs, Data Analysis, Pandas, NumPy

Tools & Methods: Agile/Scrum, SDLC, Jira, DBeaver, Jupyter Notebooks, Tableau, Wireshark, VS Code